

Splitting method and QSample

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April 27, 2026

Abstract

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0.1 Splitting method overview

The splitting method is based on an estimation method to calculate the free energy difference between two highly overlapping canonical ensembles[?]. Consider a system with a configuration space defined by coordinates $\{q_1, \dots, q_N\}$, on which we define a temperature-scaled energy function $U_0(q_1, \dots, q_N) = \Phi_0(q_1, \dots, q_N)/k_B T$. With this energy, we can build the partition function for the ensemble:

$$Z_0 = \int \exp[-U_0(q_1, \dots, q_N)] dq_1 \dots dq_N$$

An exact calculation of Z_0 is often intractable, but given the same configuration space, the ratio between two partition functions Z_0/Z_1 defined by energy functions U_0 and U_1 is tractable by Monte Carlo methods.

To calculate this ratio, we define a Metropolis function $M(x) = \min\{1, e^{-x}\}$, and notice that, for a given configuration of the ensemble, the two energy functions have the property

$$M(U_1 - U_0) \exp(-U_0) = M(U_0 - U_1) \exp(-U_1).$$

We can then integrate both sides over the configuration space and obtain:

$$\begin{aligned} \int M(U_1 - U_0) \exp(-U_0) dq_1 \dots dq_N &= \int M(U_0 - U_1) \exp(-U_1) dq_1 \dots dq_N = \\ Z_0 \frac{\int M(U_1 - U_0) \exp(-U_0) dq_1 \dots dq_N}{Z_0} &= Z_1 \frac{\int M(U_0 - U_1) \exp(-U_1) dq_1 \dots dq_N}{Z_1} = \\ Z_0 \langle M(U_1 - U_0) \rangle_0 &= Z_1 \langle M(U_0 - U_1) \rangle_1 \end{aligned} \quad (1)$$

So that we obtain the ratio as:

$$\frac{Z_0}{Z_1} = \frac{\langle M(U_0 - U_1) \rangle_1}{\langle M(U_1 - U_0) \rangle_0}. \quad (2)$$

The quantities on the right-hand side can be calculated by direct Monte Carlo sampling if $|U_1 - U_0|$ is not too big. One way around if this difference is big is to define intermediate energy functions U_i to create a chain of highly overlapping ensembles between the two target ensembles. This is especially simple to do in our particular case of estimating the error rate of QEC codes, as we will see below.

0.1.1 Splitting method for estimating QEC thresholds

An application of the splitting method for the calculation of quantum error correction thresholds can be found in [?]. In the calculation of QEC thresholds, each energy function corresponds to a physical error rate p for which we want to find the logical error rate $P(p)$. We begin with a high enough p_0 for which we can use regular Monte Carlo methods to obtain a reliable estimate $P(p_0)$, and obtain $P(p)$ by multiplying over a chain of t physical error rates $p_0 > p_1 > \dots > p_{t-1} > p$:

$$P(p) = P(p_0) \cdot \frac{P(p_1)}{P(p_0)} \cdot \dots \cdot \frac{P(p)}{P(p_{t-1})}$$

If the circuit we are testing has N possible error locations, then a bitstring E of errors has a probability $\pi_j = p_j^{|E|} (1 - p_j)^{N - |E|}$ to occur. Denoting the set of all bitstrings that lead to a logical fault by $\mathcal{F}$, the probability of a bitstring E given the presence of a logical fault is $\pi_j(E|\mathcal{F}) = \pi_j(E)/P(p_j)$. Then, any function of error configurations can be defined by:

$$\langle f \rangle_j = \sum_{E \in \mathcal{F}} \pi_j(E|\mathcal{F}) f(E)$$

Which we can estimate using T Monte Carlo samples by

$$\langle \hat{f} \rangle_j = \frac{1}{T} \sum_{\alpha=1}^T T f(E_\alpha)$$

provided that the distribution of E_α approaches $\pi_j(E|\mathcal{F})$. We can obtain such a set $\{E_\alpha\}_j$ by performing metropolis sampling over error configurations, beginning with a bitstring E_0 that is known to cause a logical fault, and given a last known failing configuration E_i , adding a new bistring E_{i+1} that differs from E_i on a single error location with probability $q = \min 1, \frac{\pi_j(E_i)}{\pi_j(E_{i+1})}$. The E_i 's will form a Markov chain that will eventually become a stationary distribution on $\mathcal{F}$ that approximates $\pi_j(E|\mathcal{F})$.

With two Markov chains $\mathcal{M}_{j-1}$ and $\mathcal{M}_j$ we can calculate the ratio $r_j = \frac{P(p_j)}{P(p_{j-1})}$ by:

$$r_j = \frac{P(p_j)}{P(p_{j-1})} = c \cdot \frac{\langle g(c\pi_{j-1}/\pi_j) \rangle_{j-1}}{\langle g(c^{-1}\pi_j/\pi_{j-1}) \rangle_j}$$

for some constant c whose optimal value is r_j , which typically we arrive at by iterating the equation above 3 times. The algorithm is detailed below.

0.2 The Splitting method

Algorithm 1 Splitting Method for Estimating $P(p)$

1: **Input:** Probability sequence $p_1, \dots, p_t = p$, initial chain length T_{init} , initial failure probability $P(p_1)$.
2: **Parameters:** Scaling factor $\lambda = 2$, target relative error $\epsilon = 0.25$, function $g(x) = \frac{1}{1+x}$.
3: **Output:** Estimated failure probability $\hat{P}(p)$ with relative error $< \epsilon$.
4: Initialize $\hat{P} \leftarrow P(p_1)$.
5: **for** $j = 1$ to t **do**
 Gather Metropolis samples until desired precision reached
6: Initialize Markov chain $\mathcal{M}_j = [E_0]$, where E_0 is a known failing fault configuration.
7: Set sequence length $T_j \leftarrow T_{\text{init}}$
8: Initialize $\lambda' \leftarrow \lambda$
9: **while** $|\mathcal{M}_j| < T_j + 1$ **do**
10: Extend $\mathcal{M}_j$ using Metropolis updates (Algorithm 3 with p_j) until length $T_j + 1$: $\mathcal{M}_j = [E_0, E_1, \dots, E_{T_j}]$
11: Compute sample estimates: $\langle \hat{g}_{\pm} \rangle_j = \frac{1}{T_j} \sum_{\alpha=1}^{T_j} g\left(\frac{\pi_j(E_{\alpha})}{\pi_{j\pm 1}(E_{\alpha})}\right)$
12: Compute sample variances: $s_{\pm}^2 = \frac{1}{T_j-1} \sum_{\alpha=1}^{T_j} \left(g\left(\frac{\pi_j(E_{\alpha})}{\pi_{j\pm 1}(E_{\alpha})}\right) - \langle \hat{g}_{\pm} \rangle_j\right)^2$
13: Estimate statistical error: $\sigma = \max\left(\frac{s_+}{\langle \hat{g}_+ \rangle_j}, \frac{s_-}{\langle \hat{g}_- \rangle_j}\right) \frac{1}{\sqrt{T_j}}$
14: Compute sample subset estimates (first half): $\langle \hat{g}'_{\pm} \rangle_j = \frac{1}{\lfloor T_j/2 \rfloor} \sum_{\alpha=1}^{\lfloor T_j/2 \rfloor} g\left(\frac{\pi_j(E_{\alpha})}{\pi_{j\pm 1}(E_{\alpha})}\right)$
15: Estimate mixing discrepancy: $\Delta = \max\left(\frac{|\langle \hat{g}_+ \rangle_j - \langle \hat{g}'_+ \rangle_j|}{\langle \hat{g}_+ \rangle_j}, \frac{|\langle \hat{g}_- \rangle_j - \langle \hat{g}'_- \rangle_j|}{\langle \hat{g}_- \rangle_j}\right)$
16: **if** $(\sigma + \Delta) > \epsilon/\sqrt{t}$ **then**
17: $T_j \leftarrow T_j + \lceil \lambda' \cdot T_{\text{init}} \rceil$
18: $\lambda' \leftarrow \lambda \cdot \lambda'$
19: **end if**
20: **end while**
21: **Estimate ratio using Metropolis samples from current and previous round**
22: **if** $j \geq 2$ **then**
23: Initialize $c \leftarrow 1$
24: **for** $i \in \{1, 2, 3\}$ **do**
25: Compute ratio estimate:
$$\hat{r}_{j-1} = c \cdot \frac{\frac{1}{T_{j-1}} \sum_{\alpha=1}^{T_{j-1}} g\left(\frac{c \pi_{j-1}(E'_{\alpha})}{\pi_j(E'_{\alpha})}\right)}{\frac{1}{T_j} \sum_{\alpha=1}^{T_j} g\left(\frac{\pi_j(E_{\alpha})}{c \pi_{j-1}(E_{\alpha})}\right)}$$

26: $c \leftarrow \hat{r}_{j-1}$
27: **end for**
28: Update failure probability estimate:
$$\hat{P} \leftarrow \hat{P} \cdot \hat{r}_{j-1}$$

29: **end if**
30: **end for**
31: **return** $\hat{P}$

0.3 Results

Comparison with qsample for distance 3 rotated surface code without any rounds of error correction:

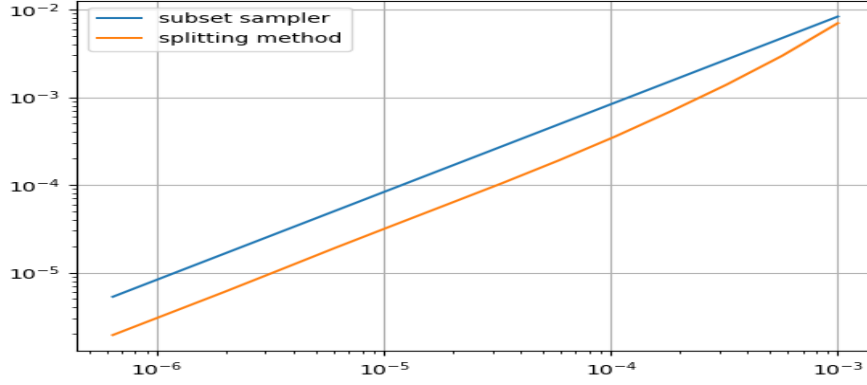


Figure 1: Comparison of logical error rates obtained by QSample and the splitting method for rotated distance 3 surface code.

Comparison with qsample for repetition code for different values of T_{init} :

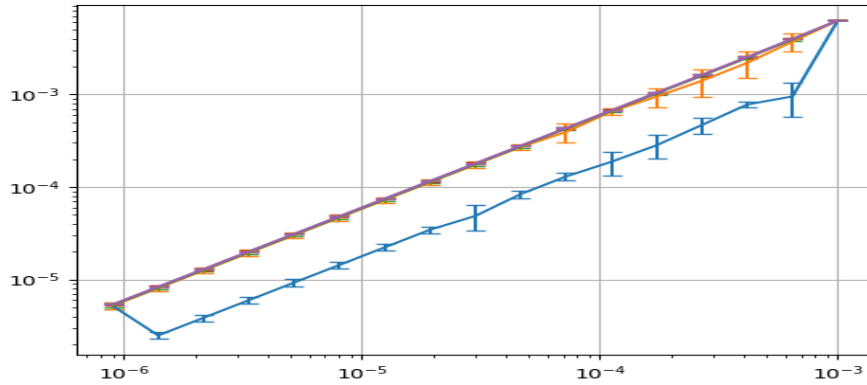


Figure 2: Comparison of logical error rates obtained by splitting method for different initial Markov chain lengths.

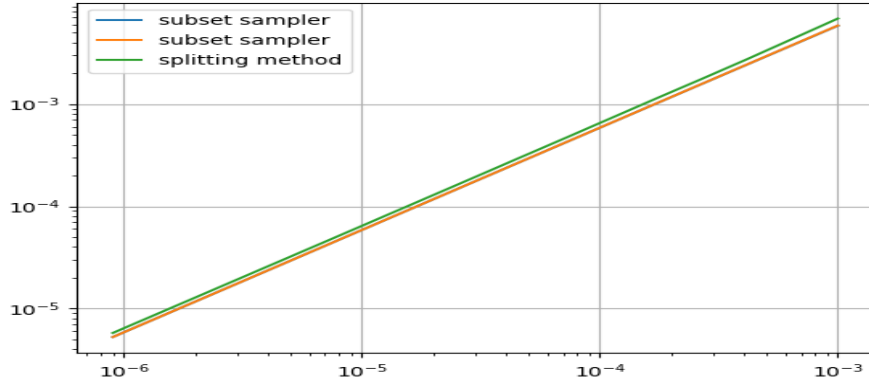


Figure 3: Comparison of logical error rates obtained by QSample and the splitting method for repetition code.

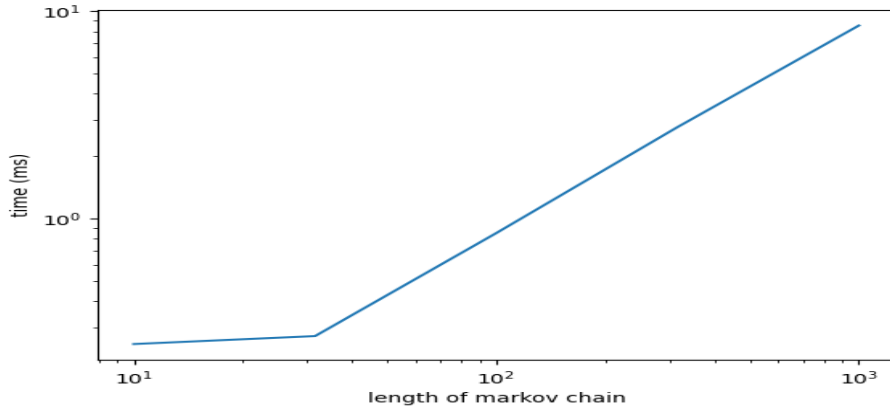


Figure 4: Computing time of splitting method for different initial Markov chain lengths

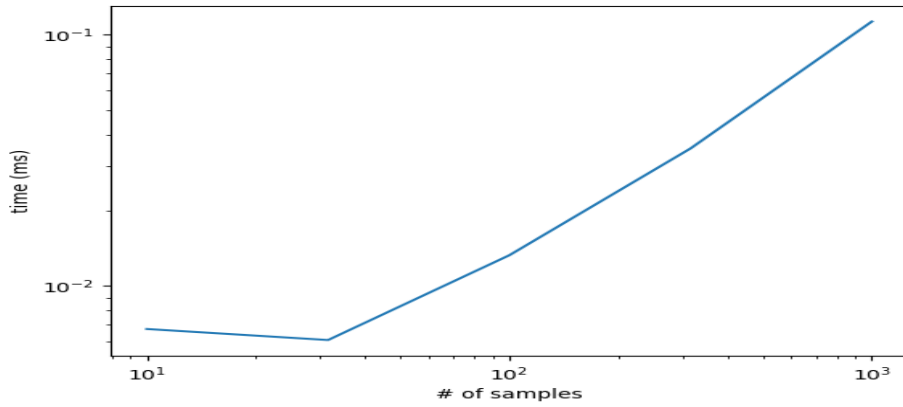


Figure 5: Computing time of qsample for different numbers of samples

0.4 Conclusion

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